

U.S. DEPARTMENT OF COMMERCE
NATIONAL BUREAU OF STANDARDS
WASHINGTON 25, D.C.

NATIONAL BUREAU OF STANDARDS

REPORT OF TEST

2.2/179390-1

on

FAIRCHILD CARTOGRAPHIC CAMERA TYPE T-12

No. 33-118

Equipped with

Bausch & Lomb Planigon Lens No. XF6800

Submitted by

Abrams Aerial Survey Corporation
Lansing 1, Michigan

The lens contained in this camera has a nominal focal length of 6 inches and maximum aperture of $f/6.3$. All measurements were made at aperture $f/8$. These measurements were made with collimated incident light, using a K-3 filter, a tungsten source and Eastman Kodak spectroscopic emulsion Type V-F and Aero-graphic Plus-X on micro flat glass plates. Development was in D19 at 68°F for three minutes with continuous agitation.

I. Focal Lengths

Equivalent focal length	152.21 mm
Calibrated focal length	152.20 mm

The probable errors of these determinations of focal length do not exceed ± 0.10 mm.

II. Distortion**1. Distortion referred to the equivalent focal length**

	7.5°	15°	22.5°	30°	37.5°	45°
Average	0	-4	-10	-15	2	4

2. Distortion referred to the calibrated focal length

Radius	7.5°	15°	22.5°	30°	37.5°	45°
1	1	-3	-6	-16	-6	9
2	1	4	1	-1	20	21
3	1	-4	-11	-12	8	9
4	1	-4	-13	-15	0	4
Average	1	-2	-7	-11	6	11

Radii 1 and 3 are on the same diagonal but separated in azimuth by 180°. Radii 2 and 4 are on the diagonal making an angle of 90° with 1 and 3. The values of the distortion are measured in microns and indicate the displacement of the image from its distortion-free position. The probable error does not exceed ±10 microns.

III. Resolving Power

Emulsion	0°	7.5°	15°	22.5°	30°	37.5°	45°
V-F	53	63	63	53	32	19	23
Tangential							
Radial	53	63	63	53	46	39	39
Plus-X							
Tangential	39	46	46	39	23	19	19
Radial	39	46	46	39	32	27	32

The values of the resolving power are given at 7.5° intervals from the center of the field and are obtained by photographing suitable test charts comprised of patterns of parallel lines. The series of patterns of the test chart are imaged on the negative with the lines spaced in a geometric series of the fourth root of two lines to the millimeter. The row marked "tangential" gives the number of lines per millimeter in the image on the negative of the finest pattern of the test chart that is distinctly resolved into separate lines when the lines lie perpendicular to the radius drawn from the center of the field. The row marked "radial" gives similar values for the pattern of test lines lying parallel to the radius.

IV. Principal Point

The lines joining opposite pairs of collimation index markers intersect at an angle of $90^\circ \pm 1$ minute, and their intersection indicates the location of the principal point, with a probable error not exceeding ± 0.03 mm.

V. Collimation Marker Separation

A - B	235.20 mm
C - D	235.23 mm

Markers A and B lie in the line of flight. The probable errors in these separations do not exceed ± 0.02 mm.

VI. Tangential Distortion

0"	$\pm 2.5^\circ$	$\pm 30^\circ$	$\pm 37.5^\circ$	$\pm 45^\circ$
0	0	1	2	3

The values of the tangential distortion are measured in microns and indicate the displacement of the image from its distortion-free position. These values represent a displacement of the central image from a straight line connecting corresponding image points at equal but opposite angular separations from the axis. The probable error does not exceed ± 5 microns.

The two surfaces of the filter accompanying this camera are parallel to within ten seconds of arc.

VII. Calibrated Focal Length Marker Separation

A - Side	152.22 mm
B - Side	152.22 mm
C - Side	152.22 mm
D - Side	152.22 mm

The probable errors in these separations do not exceed ± 0.02 mm.

VIII. Shutter Calibration

Indicated Shutter Setting	Effective Shutter Speed Seconds	Efficiency Percent
1/150	.0070 = 1/143	93%
1/300	.0034 = 1/296	90%
1/500	.0018 = 1/552	85%

The effective shutter speeds were determined with the lens at aperture $f/8$ and are correct within $\pm 3\%$. The technique used was a modification of the method described in American Standard PH 3-4-1959.

For the Director,

Francis E. Washer, Chief
Refractometry Section
Metrology Division

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WPT:sten:shf